



main.py

Visualize

Run

Output

```
1 def dice_throw(num_dice, faces, target_sum):
2     ways = [[0] * (target_sum + 1) for _ in range
3             (num_dice + 1)]
4     ways[0][0] = 1
5     for i in range(1, num_dice + 1):
6         for j in range(1, target_sum + 1):
7             total = 0
8             for k in range(1, faces + 1):
9                 if j - k >= 0:
10                    total = total + ways[i - 1][j - k]
11            ways[i][j] = total
12    return ways[num_dice][target_sum]
13
14 num_dice = 3
15 faces = 6
16 target_sum = 8
17
18 result = dice_throw(num_dice, faces, target_sum)
19 print("Number of ways =", result)
```

Number of ways = 21

=== Code Execution Successful ===



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Output

```
1
2 def dice_throw(num_dice, faces, target_sum):
3     ways = [[0] * (target_sum + 1) for _ in range
4               (num_dice + 1)]
5     ways[0][0] = 1
6     for i in range(1, num_dice + 1):
7         for j in range(1, target_sum + 1):
8             total = 0
9             for k in range(1, faces + 1):
10                if j - k >= 0:
11                    total = total + ways[i - 1][j - k]
12                ways[i][j] = total
13     return ways[num_dice][target_sum]
14
15 num_dice = 2
16 faces = 6
17 target_sum = 7
18
19 result = dice_throw(num_dice, faces, target_sum)
20 print("Number of ways =", result)
```

Number of ways = 6

=== Code Execution Successful ===



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Run

Output



JS

TS



```
1 def dice_throw(num_dice, faces, target_sum):
2     ways = [[0] * (target_sum + 1) for _ in range
3             (num_dice + 1)]
4     ways[0][0] = 1
5     for i in range(1, num_dice + 1):
6         for j in range(1, target_sum + 1):
7             total = 0
8             for k in range(1, faces + 1):
9                 if j - k >= 0:
10                    total = total + ways[i - 1][j - k]
11            ways[i][j] = total
12    return ways[num_dice][target_sum]
13
14 num_dice = 4
15 faces = 4
16 target_sum = 10
17
18 result = dice_throw(num_dice, faces, target_sum)
19 print("Number of ways =", result)
20
21 # Expected Output: 44
22
23
```

Number of ways = 44

=== Code Execution Successful ===



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Visualize

Run

Output

```
1 def dice_throw(num_dice, faces, target_sum):
2     ways = [[0] * (target_sum + 1) for _ in range
3             (num_dice + 1)]
4     ways[0][0] = 1
5     for i in range(1, num_dice + 1):
6         for j in range(1, target_sum + 1):
7             total = 0
8             for k in range(1, faces + 1):
9                 if j - k >= 0:
10                    total = total + ways[i - 1][j - k]
11            ways[i][j] = total
12    return ways[num_dice][target_sum]
13
14 num_dice = 3
15 faces = 6
16 target_sum = 12
17
18 result = dice_throw(num_dice, faces, target_sum)
19 print("Number of ways =", result)
20
21 # Expected Output: 25
22
```

Number of ways = 25

=== Code Execution Successful ===



main.py

Visualize

Run

Output

```
1
2 def dice_throw(num_dice, faces, target_sum):
3     ways = [[0] * (target_sum + 1) for _ in range
4             (num_dice + 1)]
5     ways[0][0] = 1
6     for i in range(1, num_dice + 1):
7         for j in range(1, target_sum + 1):
8             total = 0
9             for k in range(1, faces + 1):
10                if j - k >= 0:
11                    total = total + ways[i - 1][j - k]
12                ways[i][j] = total
13     return ways[num_dice][target_sum]
14
15 num_dice = 5
16 faces = 6
17 target_sum = 15
18
19 result = dice_throw(num_dice, faces, target_sum)
20 print("Number of ways =", result)
21
```

Number of ways = 651

=== Code Execution Successful ===



main.py

Visualize

Run

Output

```
1 def dice_throw(num_dice, faces, target_sum):
2     ways = [[0] * (target_sum + 1) for _ in range
3             (num_dice + 1)]
4     ways[0][0] = 1
5     for i in range(1, num_dice + 1):
6         for j in range(1, target_sum + 1):
7             total = 0
8             for k in range(1, faces + 1):
9                 if j - k >= 0:
10                    total = total + ways[i - 1][j - k]
11            ways[i][j] = total
12    return ways[num_dice][target_sum]
13
14 num_dice = 3
15 faces = 5
16 target_sum = 9
17
18 result = dice_throw(num_dice, faces, target_sum)
19 print("Number of ways =", result)
```

Number of ways = 19

=== Code Execution Successful ===



main.py

Visualize

Run

Output

```
1 def dice_throw(num_dice, faces, target_sum):
2     ways = [[0] * (target_sum + 1) for _ in range
3             (num_dice + 1)]
4     ways[0][0] = 1
5     for i in range(1, num_dice + 1):
6         for j in range(1, target_sum + 1):
7             total = 0
8             for k in range(1, faces + 1):
9                 if j - k >= 0:
10                    total = total + ways[i - 1][j - k]
11            ways[i][j] = total
12    return ways[num_dice][target_sum]
13
14 num_dice = 2
15 faces = 10
16 target_sum = 11
17
18 result = dice_throw(num_dice, faces, target_sum)
19 print("Number of ways =", result)
```

Number of ways = 10

=== Code Execution Successful ===



main.py

Visualize

Run

Output

```
1
2 def dice_throw(num_dice, faces, target_sum):
3     ways = [[0] * (target_sum + 1) for _ in range
4               (num_dice + 1)]
5     ways[0][0] = 1
6     for i in range(1, num_dice + 1):
7         for j in range(1, target_sum + 1):
8             total = 0
9             for k in range(1, faces + 1):
10                if j - k >= 0:
11                    total = total + ways[i - 1][j - k]
12                ways[i][j] = total
13     return ways[num_dice][target_sum]
14
15 num_dice = 2
16 faces = 6
17 target_sum = 7
18
19 result = dice_throw(num_dice, faces, target_sum)
20 print("Number of ways =", result)
21
```

Number of ways = 10

=== Code Execution Successful ===



main.py

Visualize

Run

Output

```
1 def dice_throw(num_dice, faces, target_sum):
2     ways = [[0] * (target_sum + 1) for _ in range
3             (num_dice + 1)]
4     ways[0][0] = 1
5     for i in range(1, num_dice + 1):
6         for j in range(1, target_sum + 1):
7             total = 0
8             for k in range(1, faces + 1):
9                 if j - k >= 0:
10                    total = total + ways[i - 1][j - k]
11            ways[i][j] = total
12    return ways[num_dice][target_sum]
13
14 num_dice = 3
15 faces = 6
16 target_sum = 8
17
18 result = dice_throw(num_dice, faces, target_sum)
19 print("Number of ways =", result)
20
```

Number of ways = 21

=== Code Execution Successful ===



main.py

Visualize

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Output

```
1 def dice_throw(num_dice, faces, target_sum):
2     ways = [[0] * (target_sum + 1) for _ in range
3             (num_dice + 1)]
4     ways[0][0] = 1
5     for i in range(1, num_dice + 1):
6         for j in range(1, target_sum + 1):
7             total = 0
8             for k in range(1, faces + 1):
9                 if j - k >= 0:
10                    total = total + ways[i - 1][j - k]
11            ways[i][j] = total
12    return ways[num_dice][target_sum]
13
14 num_dice = 4
15 faces = 6
16 target_sum = 10
17
18 result = dice_throw(num_dice, faces, target_sum)
19 print("Number of ways =", result)
```

Number of ways = 80

=== Code Execution Successful ===

JS

TS

GO





main.py

Visualize

Run

Output

```
1 def dice_throw(num_dice, faces, target_sum):
2     ways = [[0] * (target_sum + 1) for _ in range
3             (num_dice + 1)]
4     ways[0][0] = 1
5     for i in range(1, num_dice + 1):
6         for j in range(1, target_sum + 1):
7             total = 0
8             for k in range(1, faces + 1):
9                 if j - k >= 0:
10                    total = total + ways[i - 1][j - k]
11            ways[i][j] = total
12    return ways[num_dice][target_sum]
13
14 num_dice = 2
15 faces = 4
16 target_sum = 5
17
18 result = dice_throw(num_dice, faces, target_sum)
19 print("Number of ways =", result)
20
```

Number of ways = 4

=== Code Execution Successful ===



JS

TS

